

Section 7 Programming Challenge Problems

#1: Mirrored Smileys

First, create a mashup of the [ClickAndSmile.py](#) and [ClickKaleidoscope.py](#) programs to draw a smiley face in four mirrored corners of the screen when you click, just like the kaleidoscope program did with spirals. If you'd like an advanced challenge, draw two of the smiley faces flipped upside down so that they really look mirrored across the x-axis from top to bottom.

#2: PingPongChallenge

Second, modify the Ping-Pong calculator so it asks the user for a number of Ping-Pong balls as input and tells the user how tall those Ping-Pong balls would be if stacked on top of one another, and how much that number of Ping-Pong balls would weigh.

#3: ArrowDrawChallenge

Change the [ArrowDraw.py](#) program to allow the user to turn the turtle in smaller increments—say 45 degrees (or even 30, or 15)—to give them finer control of the turtle. Then, add more key options, like allowing the user to press the greater-than symbol (>) to make the drawing length longer, the less-than symbol (<) to shorten the drawing length, the W key to make the pen wider, and the T key to make the pen thinner. To make it a great drawing program, add feedback in the form of drawing a string on the screen to show the pen width, segment length, and turtle's direction after every change.

*For sample answers to these programming challenges,
go to <http://www.TeachYourKidsToCode.com>*